AirConnect India

Machine Learning-Based Analysis of Indian Airport Passenger Traffic and Connectivity

K-Means · Random Forest regression · Offline research interface

Group members: [Names / Register numbers / University emails]

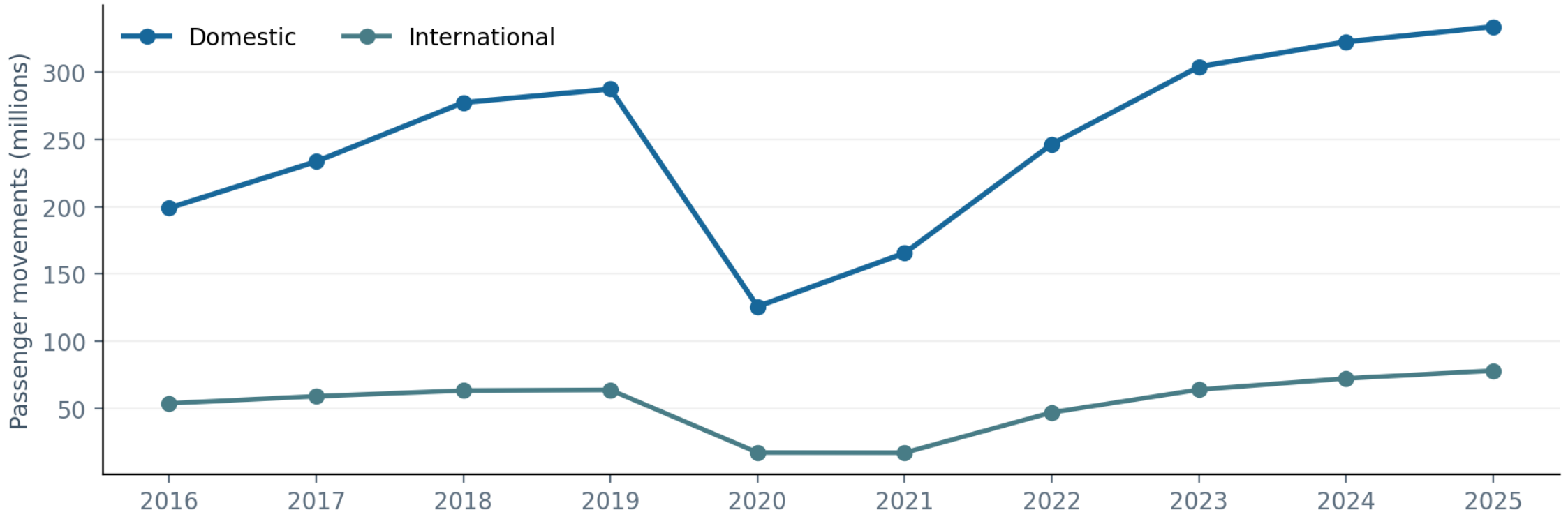
Problem and motivation

- India: 74 operational facilities in 2014; 165 as of July 2026.
- Infrastructure growth ≠ number of airports reporting passenger movements.
- Does traffic growth appear balanced? What patterns separate airports?
- Can a simple model beat a previous-year prediction?

Data collection and preprocessing

- DGCA-derived national passenger movements: 2016–2025.
- Airport-level source rows: 614, covering 2022–2025.
- Arrival + departure movements; not unique passenger counts.
- Validate keys and totals; pivot airport/year; transform numeric features.

National traffic patterns

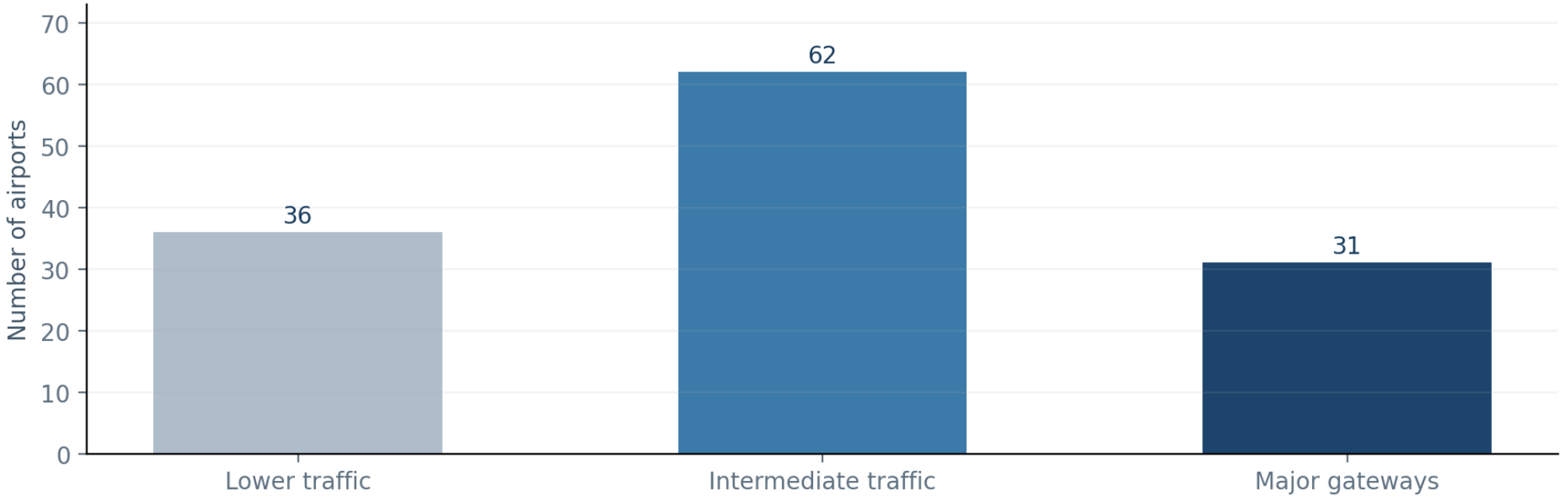


2016–2025 domestic passenger movements: +67.8%. A sharp decline is visible in 2020.

Machine learning workflow

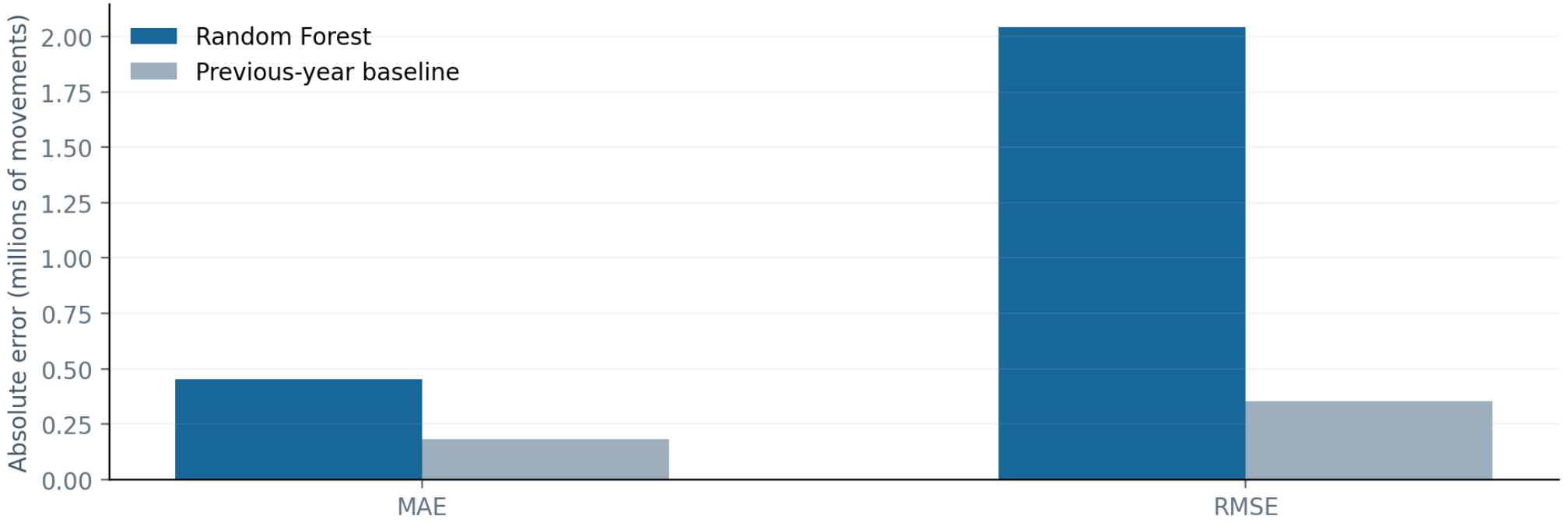
- Input CSV → validation → airport-year features.
- K-Means: log traffic features → standard scaling → 3 clusters.
- Random Forest: lag-1 and lag-2 domestic traffic → annual target.
- Chronological holdout: 2024 train, 2025 test.

K-Means results



129 airports | k = 3 | Silhouette Score = 0.6235

Random Forest evaluation



Test 2025: Random Forest MAE 451,552 | Prior-year baseline MAE 182,476

Website demonstration

- Offline HTML / CSS / JavaScript; no login, server or cloud.
- Overview: aviation growth indicators and historical traffic chart.
- Explorer: India map, year filter, airport search, CSV export.
- ML: cluster groups, prediction comparison and method definitions.
- Methodology: reproducible steps, definitions and source citations.

Findings and limitations

- Observed domestic and international traffic increased over the period.
- K-Means distinguishes substantially different traffic groups.
- Prior-year forecast outperforms this Random Forest on 2025 holdout.
- Low traffic ≠ proven underutilisation; international activity ≠ route permission.

Conclusion and references

- Airport traffic is measurable, but capacity utilisation needs other data.
- Simpler forecast models can outperform complex alternatives.
- A transparent, reproducible offline interface supports evaluation.

Sources: DGCA-derived India Aviation Stats (Kubicek, 2026); Ministry of Civil Aviation; airport coordinates (Himpfen); Natural Earth.